

ROSHNI

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Education

- **Sahyadri College of Engineering and Management**
B.E in Computer Science and Engineering (CSE)

Mangaluru, India
December 2021 - May 2025

Skills Summary

- **Languages:** Python, SQL, HTML, CSS
- **Frameworks / Libraries:** FastAPI, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Matplotlib, Seaborn, OpenCV, OpenAI API
- **Tools:** Docker, Tableau, Excel
- **Core Competencies:** Data Analysis, Machine Learning, LLM-based Applications, Visualization, Problem-Solving

Professional Experience

Logic Hive Solutions Pvt. Ltd: Technology Intern

February 2025 – May 2025

- Built an LLM-powered chatbot using the OpenAI API, enabling smoother client interactions.
- Designed Qdrant (VectorDB) and MongoDB pipelines for semantic search and fast retrieval, enhancing response accuracy.
- Implemented CRUD features within the chatbot, improving usability and overall client experience.
- Tech: Python, LLM, FastAPI, Docker, MongoDB, VectorDB

Projects

Cancer Gene Expression Classifier

- Developed a machine learning pipeline to classify 5 cancer subtypes from 801 RNA-seq gene expression profiles, achieving **99.2% accuracy** and **0.993 Macro F1-score** using XGBoost with Optuna-based hyperparameter optimization.
- Applied ANOVA F-score feature selection to reduce **20,531 genes** to the **top 500 predictive biomarkers** and leveraged SHAP explainability to identify key gene contributors influencing model predictions.
- Built and deployed an end-to-end workflow encompassing data preprocessing, model training, evaluation, and an interactive Streamlit dashboard, containerized using Docker for reproducible deployment.
- **Tech Stack:** Python, XGBoost, Optuna, SHAP, Scikit-learn, Pandas, NumPy, Streamlit, Docker

DocuMind AI - Intelligent Document Processing System

- Built an AI-powered document processing system to extract and summarize content from research papers and images using Tesseract OCR (PSM 6) and a local LLM (Qwen 2.5 via Ollama), ensuring high accuracy and data privacy.
- Designed a modular backend with FastAPI and an interactive UI using Streamlit, enabling seamless file uploads and real-time summarization.
- Containerized the application with Docker, optimizing performance by using a lightweight model to significantly reduce memory usage while maintaining accuracy.
- **Tech Stack:** Python, FastAPI, Streamlit, Tesseract OCR, Ollama, Qwen 2.5, Docker

Movie Recommendation System

- Built a personalized movie recommendation system using User-Based and Item-Based Collaborative Filtering on the MovieLens dataset, achieving 0.90 RMSE and 0.70 MAE.
- Developed data preprocessing and model pipelines using Pandas, NumPy, and Scikit-learn, leveraging similarity metrics for accurate recommendations.
- Created an interactive Streamlit application for real-time movie search and personalized suggestions.
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, Streamlit, SciPy

Achievements and Certifications

Deloitte Australia Data Analytics Job Simulation — Forage

June 2025

- Completed a Deloitte job simulation involving data analysis and forensic technology.
- Built an interactive data dashboard using Tableau.
- Used Excel to classify datasets and derive business insights.